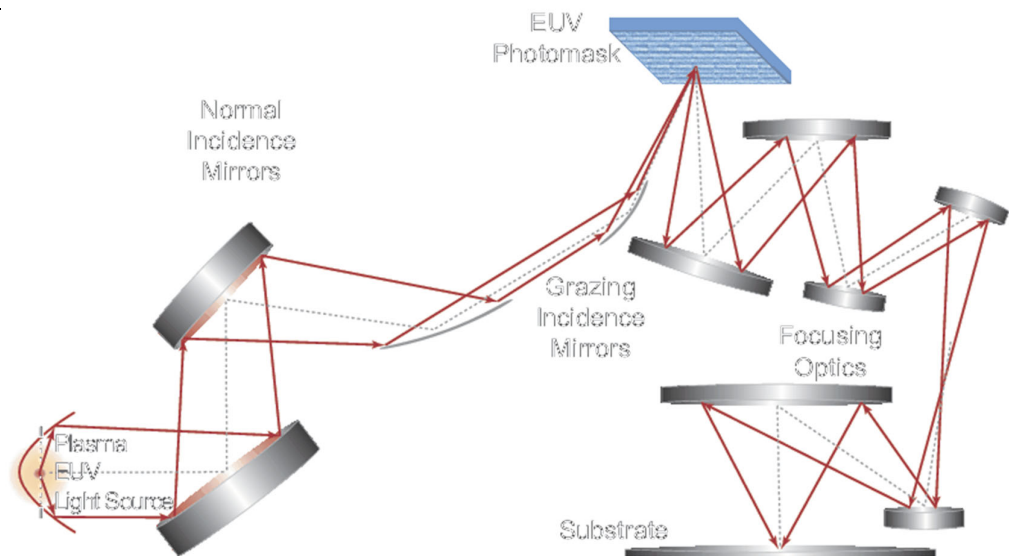


01.119/99.503: Nanoelectronics and Technology

Homework 3-Solutions

1. Explain Extreme Ultraviolet (EUV) lithography procedure with the schematic of EUV system. Briefly discuss the challenges of using EUV lithography.

Solution:



Ref: <https://www.newport.com/n/extreme-uv-photolithography>

[NOTE: For detailed information on the working of EUV lithography, refer to reading material on EUV topic]

Functional Blocks of an EUV Lithography System

Extreme Ultraviolet (EUV) lithography operates at a wavelength of 13.5 nm and requires a fully reflective, vacuum-based optical system. Its main functional blocks are:

- **EUV Light Source**
A high-power laser-produced plasma (LPP) source is used, where a CO₂ laser strikes tin (Sn) droplets to generate EUV radiation at 13.5 nm.
- **Collector Optics**
A multi-layer reflective collector mirror gathers and focuses the emitted EUV light toward the illumination system while filtering unwanted radiation and debris.
- **Illumination Optics (Condenser System)**
Shapes and homogenizes the EUV beam to provide uniform illumination of the mask with the desired numerical aperture and illumination profile.
- **EUV Mask (Reticle)**
A reflective mask consisting of Mo/Si multilayers with absorber patterns. Unlike DUV, EUV masks work in reflection and are protected by a thin EUV pellicle.
- **Projection Optics System (POS)**
A set of ultra-precise multilayer reflective mirrors that reduce and project the mask pattern onto the wafer (typically 4× reduction).
- **Wafer Stage and Photoresist**
The wafer, coated with EUV-sensitive photoresist, is scanned synchronously with the mask under ultra-high vacuum to form the patterned image.
- **Vacuum System**
The entire optical path operates in high vacuum, since EUV light is strongly absorbed by air.

Challenges of using EUV lithography:

- 13.5 nm source
- Optics system
- Special properties required for the photo mask
- Requirements of specific photo resist properties

2. Describe how the performance of the capacitor affects the performance of the DRAM cell.

Ans: DRAM uses capacitors that lose charge over time due to leakage, even if the supply voltage is maintained. Since the charge on a capacitor decays when a voltage is removed, DRAM must be supplied with a voltage to retain memory and thus the memory is volatile.

READ/WRITE times depend on the performance of the capacitor. Dynamic memory, requires periodic refresh and reading dynamic memory requiring a recharge of the storage cells in the row that has been read. If these processes are imperfect, a read operation can cause soft errors. In particular, there is a risk that some charge can leak between nearby cells, causing the refresh or read of one row to cause a *disturbance error* in an adjacent or even nearby row.

Thus, in a DRAM cell, data is stored as charge on the capacitor, so the capacitor's performance directly determines the cell's speed, reliability, and power. A higher capacitance provides more stored charge, improving sensing margin, read reliability, and retention time, while reducing soft-error probability. Leakage through the capacitor dielectric causes charge loss, which shortens retention time and increases refresh frequency and power consumption. Dielectric quality and capacitor scaling therefore strongly affect DRAM read/write performance, refresh overhead, and long-term reliability.

3. Briefly explain WRITE and READ operation of 6-T SRAM cell.

ANS:

- **Write**

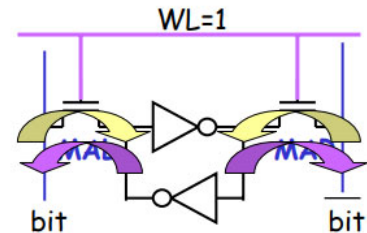
- word line = 1, access tx are ON
- new data (voltage) applied to bit and bit_bar
- data in latch overwritten with new value

- **Read**

- word line = 1, access tx are ON
- bit and bit_bar read by a sense amplifier

- **Sense Amplifier**

- basically a simple differential amplifier
- comparing the difference between bit and bit_bar
 - if bit > bit_bar, output is 1
 - if bit < bit_bar, output is 0
 - allows output to be set quickly without fully charging/discharging bit line



WRITE Operation

- **WL = HIGH** → access transistors turn ON.
- External write drivers force **BL/BL̄** to complementary values (e.g., BL = 1, BL̄ = 0 to write '1').
- The forced bitline values **override the previous latch state**, flipping the internal nodes.
- **WL = LOW** → cell is isolated and **new data is stored**.

Key point: Write requires access transistors strong enough to overpower the cell's feedback.

READ Operation

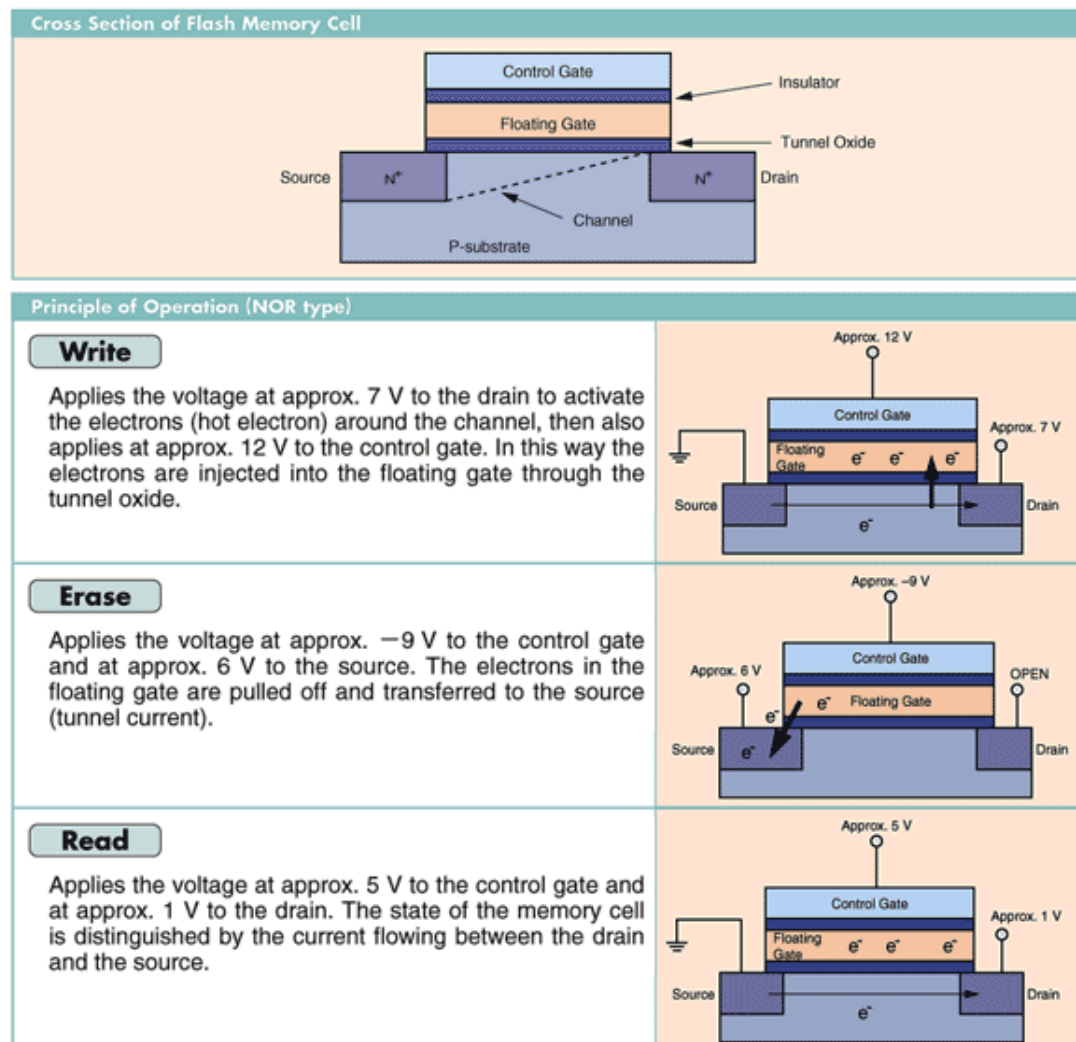
- **BL and BL̄ are precharged HIGH.**
- **WL = HIGH** → access transistors turn ON.
- Depending on stored data:
 - One internal node pulls its corresponding bitline **slightly down**.
 - The other bitline remains near HIGH.
- A **sense amplifier** detects this small voltage difference.
- **WL = LOW** → cell returns to hold state.

4. Explain the role of floating gate and tunnel oxide in flash memory cell. Describe the WRITE and READ operation in flash memory cell with the help of schematic.

Ans:

A positive charge on the control gate attracts the electrons from the channel into the floating gate, where they become trapped. The oxide layer that surrounds the floating gate keeps the electrons trapped, whether or not the flash device has power, enabling persistent storage of data bits. Hence, floating-gate has the ability to store an electrical charge for extended periods of time without a connection to a power supply.

As the thickness of floating gate decreases, the difference of charging-discharging electron density increases. In other hand, as the floating gate getting thinner, the amount of electrons involved due to PE cycle is getting higher. The electron density and amount of electrons tunnelled into the floating gate increases for thinner barrier oxide.



5. Describe the necessity of ultra-high vacuum (UHV) conditions for nanoscale characterization techniques.

Ans:

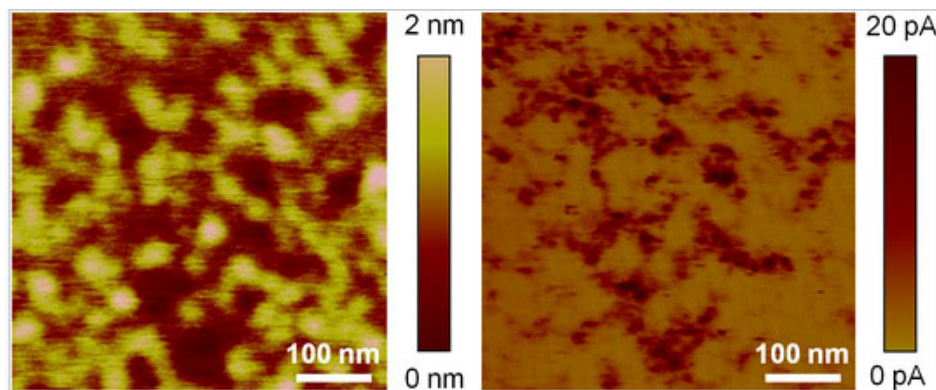
Ultra High Vacuum (UHV)

- keeping surfaces clean for hours
(surface science, epitaxial growth)
- achieving ultra high purities
- ultra pure growth
- vacuum evaporation or coating
- Avoid anodic oxidation

6. (a) Describe the key application of contact-AFM in characterizing the thin insulating materials.
- (b) Explain the working of non-contact-AFM (Amplitude modulation mode).

Ans: (a)

Conductive AFM works in contact AFM mode with conducting AFM tip. Conductive atomic force microscopy C-AFM allows us to characterize the dielectric film both topographically and electrically with nanometer resolution.

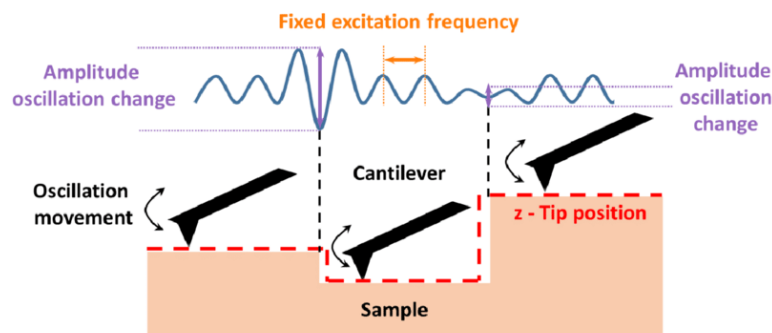


Topographic (left) and current (right) maps collected with CAFM on a polycrystalline HfO₂ stack. The images show very good spatial correlation.

(b)

The amplitude of oscillation is measured by a lock-in amplifier. The error signal is the difference of the amplitude of oscillation and the set point. The piezo-voltage, as in the constant force mode, is topography

In this mode the sensor is excited just off resonance. By exciting the sensor just above its resonant frequency, it is possible to detect forces which change the resonant frequency by monitoring the amplitude of oscillation. An attractive force on the probe causes a decrease in the sensors resonant frequency, thus the driving frequency is further from resonance and the amplitude decreases, the opposite is true for a repulsive force. The microscopes control electronics can then use amplitude as the SPM reference channel, either in feedback mode, or it can be recorded directly in constant height mode.



7. (a) Highlight the key applications of STM technique and challenges associated with STM characterization.
- (b) Draw the force-distance graph in AFM and indicate the regions where contact and non-contact AFM works.

Ans: (a)

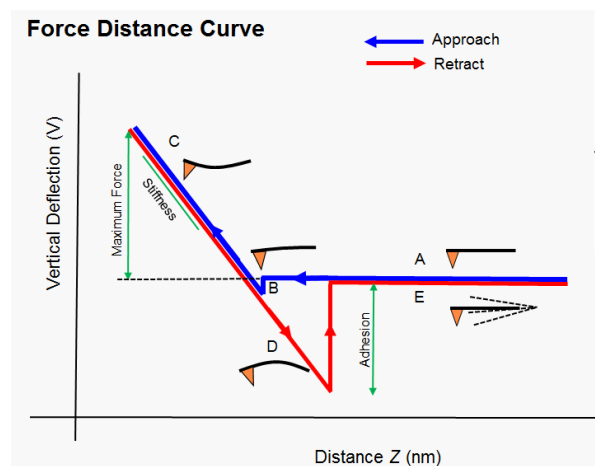
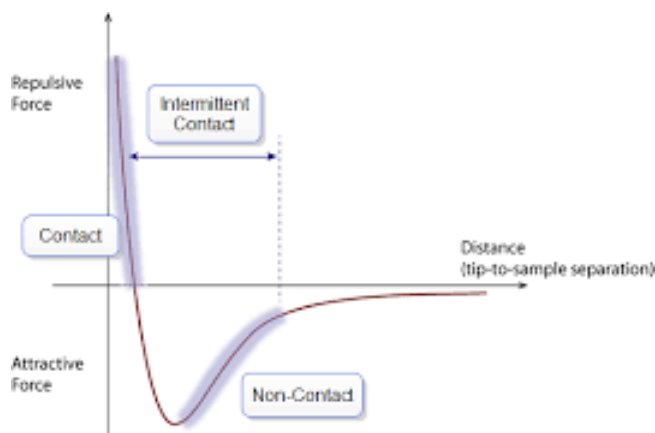
- Surface Structure: Compare to bulk structure
- Semiconductor surface structure, Nanotechnology, Superconductors, etc.
- Metal-catalyzed reactions
- Spectroscopy of single atoms
- To characterize thin dielectric materials
- Future Developments: Improve understanding of how electronic structure affects tunneling current

Challenges:

- Requires UHV conditions
- Specific sample conditions: for dielectrics, STM works for thin materials
- Sample surface must be clean
- Tip quality is critical for obtaining high resolution images

(b)

Force distance curve plays a vital role for the explanation of different types of interaction and the underlying physics for the interaction. The force is not same as in AFM is brought in or out (attraction and retraction). Force distance curve is the study of interaction force between the cluster of atoms in the AFM cantilever tip and the sample surface. What we measure in AFM is the distance between the cantilever and the sample (Z). The distance controlled during the measurement is the distance (Z) between sample surface and the rest position of the cantilever.



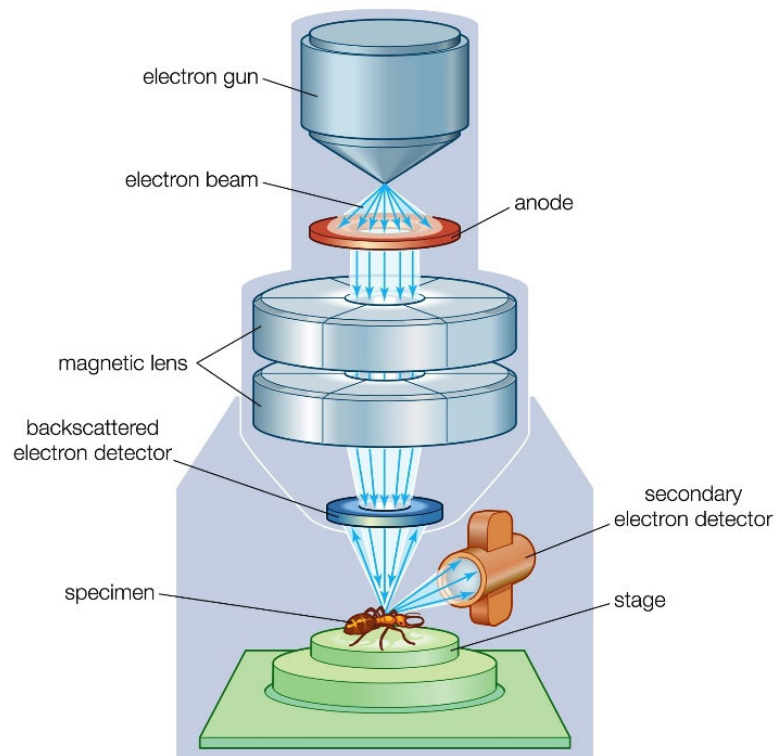
8. Explain the role of secondary electrons and backscattered electrons in characterization of materials using SEM.

Ans:

In SEM, Secondary electrons and backscattered electrons are commonly used for imaging samples: secondary electrons are most valuable for showing morphology and topography on samples and backscattered electrons are most valuable for illustrating contrasts in composition in multiphase samples.

They are a result of inelastic interactions between the primary electron beam and the sample and have lower energy than the backscattered electrons. Secondary electrons are very useful for the inspection of the topography of the sample's surface.

Back-scattered electrons (BSE) are beam electrons that are reflected from the sample by elastic scattering. Since they have much higher energy than SEs, they emerge from deeper locations within the specimen and, consequently, the resolution of BSE images is less than SE images. A backscattered electron image reveals the compositional difference in a specimen.



9. Explain the physics behind the image (micrograph) formation when electron beam passes through sample in TEM.

Ans: Transmission electron microscopes (TEM) are microscopes that use a particle beam of electrons to visualize specimens and generate a highly-magnified image. TEMs can magnify objects up to 2 million times.

TEMs employ a high voltage electron beam in order to create an image. An electron gun at the top of a TEM emits electrons that travel through the microscope's vacuum tube. The TEM employs an electromagnetic lens which focuses the electrons into a very fine beam. This beam then passes through the specimen, which is very thin, and the electrons either scatter or hit a fluorescent screen at the bottom of the microscope. An image of the specimen with its assorted parts shown in different shades according to its density appears on the screen. This image can be then studied directly within the TEM or photographed.

